

B.Sc. Semester-II Examination, 2022-23

COMPUTER SCIENCE [Honours]

Course ID : 21511

Course Code : SH/CSC/201/C-3

Course Title : Object Oriented Programming

[NEW SYLLABUS]

Time : 1 Hour 15 Minutes

Full Marks : 25

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

UNIT-I

1. Answer any five of the following questions:

1×5=5

- a) Define Constructor in Java?
- b) Define virtual function.
- c) What is a scope resolution operator?
- d) What is friend function in C++?
- e) What is polymorphism?
- f) What is java utility?
- g) What is singleton class in Java?
- h) What is a package in Java?

[Turn Over]

UNIT-II

2. Answer any **two** of the following questions: $5 \times 2 = 10$

- a) Write a C++ program to demonstrate example of Hierarchical inheritance.
- b) Explain function overloading with example.
- c) Write a Java program to find common elements between two integer arrays.
- d) What is interface in Java? Write down difference between abstract class and interface. $1+4$

UNIT-III

3. Answer any **one** of the following questions:

$10 \times 1 = 10$

- a) Discuss visibility modes in C++ with proper example of each.
- b) What is the difference between error and exception in Java? Explain exception handling mechanism in Java with example. $3+7$

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B.Sc. Semester-II Examination, 2022-23**COMPUTER SCIENCE [Honours]****Course ID : 21521****Course Code : SH/CSC/201/C-3****Course Title : Object Oriented Programming (Practical)****[NEW SYLLABUS]****Time : 2 Hours****Full Marks : 15***The figures in the right-hand margin indicate marks.**Candidates are required to give their answers in their own words as far as practicable.***Perform any **one** experiment:**

1. Write a java Program to find the sum of digits of any number of integers entered as command line arguments.
2. Write a java Program to convert given decimal number to binary number.
3. C++ Program to Generate Multiplication Table of a given number.
- ✓ 4. Write a program that show working of following functions of String and StringBuffer classes setCharAt(), setLength(), append(), insert(), concat() and equals().

[Turn Over]

5. Write a C++ Program to create a class Triangle include overloaded functions for calculating area. 2

6. C++ program to print following pattern:

```
*  
  
* *  
  
* * *  
  
* * * *  
  
* * * * *
```

7. Write a java program to create a reusable class to generate Fibonacci series.

8. C++ program to Calculate Factorial of a Number Using Recursion.

9. Write a java program that takes two numbers a and b as input, computes a/b, and invokes Arithmetic Exception to generate a message when the denominator is zero.

10. C++ Program to Add Two Matrix Using Multi-dimensional Arrays.